

Tugas: Dataset iris

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1. Praktikum Mandiri

1.1 Menghubungkan dengan Google Drive

Langkah ini digunakan agar Google Colab dapat mengakses file yang tersimpan di Google Drive. Dataset calonpembelimobil.csv yang digunakan dalam percobaan ini disimpan di dalam folder Drive. Mounting dilakukan menggunakan perintah `drive.mount("/content/drive")`.

```
1. Hubungkan ke Google Drive

[1] ✓ 35s
from google.colab import drive
drive.mount('/content/drive')

Mounted at /content/drive
```

1.2 Import Library dan Load Dataset

Library seperti pandas dan numpy digunakan untuk membaca serta memproses data. Kemudian dataset iris.csv.

```
2. Baca Dataset dari Drive

[2] ✓ 1s
import pandas as pd

# Path ke file Iris.csv di Google Drive
file_path = '/content/drive/MyDrive/Praktikum Mandiri_ML/Praktikum5/data/Iris.csv'

# Baca dataset
df = pd.read_csv(file_path)

# Tampilkan beberapa baris pertama
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Next steps: [Generate code with df](#) [New interactive sheet](#)

1.3 Pisahkan fitur (X) dan target (y).

```
3. Pisahkan Fitur dan Target

[3] # Hapus kolom Id, pisahkan fitur dan target
✓ 0s X = df.drop(['Id', 'Species'], axis=1)
    y = df['Species']
```

1.4 Bagi data training dan testing (80% : 20%)

```
4. Bagi Data Training dan Testing (80% : 20%)

[4] from sklearn.model_selection import train_test_split
2s
    # Bagi data
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )
```

1.5 Buat model Decision Tree

```
5. Buat model Decision Tree

[5] from sklearn.tree import DecisionTreeClassifier
✓ 0s

    # Inisialisasi model
    model = DecisionTreeClassifier(random_state=42)

    # Latih model
    model.fit(X_train, y_train)
```

DecisionTreeClassifier ⓘ ?

DecisionTreeClassifier(random_state=42)

1.6 Prediksi data testing

```
6. Prediksi data testing

[6]
✓ 0s
# Prediksi data test
y_pred = model.predict(X_test)
```

1.7 Evaluasi Model

```
[7]
✓ 0s
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Akurasi
accuracy = accuracy_score(y_test, y_pred)
print("Akurasi model:", accuracy)

# Laporan klasifikasi
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Confusion Matrix
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

Akurasi model: 1.0

Classification Report:

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	10
Iris-versicolor	1.00	1.00	1.00	9
Iris-virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Confusion Matrix:

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

1.8 Visualisasi Pohon

8. Visualisasi Pohon

```
[8] from sklearn import tree
    import matplotlib.pyplot as plt

    plt.figure(figsize=(12,8))
    tree.plot_tree(model, filled=True, feature_names=X.columns, class_names=y.unique())
    plt.show()
```

